浙江大学实验报告

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课程名称：Signals and Systems　任课老师：Prof. Xiaoming Chen

实验名称：Lecture3: Laplace, Z-Transform　实验日期：2026.4.7

1 Experimental Objectives

(1) Draw Continuous-Time Pole-Zero Diagrams: To learn how to use MATLAB functions (such as roots, plot, and pzplot) to find and draw pole-zero diagrams for continuous-time systems. Also, to understand how to find the Region of Convergence (ROC) for a stable system based on its poles.

(2) Draw Discrete-Time Pole-Zero Diagrams: To learn how to calculate and draw pole-zero diagrams for discrete-time systems using MATLAB, especially using the dpzplot function. To learn how to handle the situation when the numerator and denominator have different orders.

(3) Implement Noncausal Continuous-Time Filters: To learn how to use impulse and lsim functions to simulate the response of LTI systems. To master the "time-reversal" method to calculate the output of anticausal systems. Also, to use the residue function to break a complex system into causal and anticausal parts for parallel realization.

2 Experimental Principles

(1) The zeros and poles of a continuous-time system function H(s) can be found by setting its numerator and denominator to zero, respectively (using the roots function). Note that systems with the same rational expression may have different properties due to different Regions of Convergence (ROC), and the ROC of a stable LTI system must include the $j\omega\ axis$.

(2) For discrete-time system functions H(z), when calculating roots of polynomials ordered by z^{-1}, if the numerator and denominator have different orders, you must append zeros to the lower-order vector to make their lengths equal. This prevents missing poles or zeros at the origin $z = 0$.

(3) When implementing noncausal continuous-time filters, since built-in functions only simulate causal systems, you must use a "time-reversal" technique for anticausal systems: simulating the time-reversed equation and then flipping the result. For noncausal systems, you need to use partial fraction expansion (like the residue function) to split the system into causal and anticausal parts, calculate them in parallel, and add them together.

3 Experimental Contents, Results, and Analysis

3.1 Problem 1

3.1.a

Problem 1

(a). Each of the following system functions corresponds to a stable LTI system. Use **roots** to find the poles and zeros of each system function and make an appropriately labeled pole-zero diagram using **plot** as shown above.

(i) $H(s) = \dfrac{s+5}{s^2+2s+3}$

(ii) $H(s) = \dfrac{2s^2+5s+12}{s^2+2s+10}$

(iii) $H(s) = \dfrac{2s^2+5s+12}{(s^2+2s+10)(s+2)}$

Several different signals can have the same rational expression for their Laplace transform while having different regions of convergence. For example the causal and anticausal LTI systems with impulse responses

$$h_c(t) = e^{-\alpha t}u(t), \qquad h_{ac}(t) = -e^{-\alpha t}u(-t),$$

(a). Each of the following system functions corresponds to a stable LTI system. Use **roots** to find the poles and zeros of each system function and make an appropriately labeled pole-zero diagram using **plot** as shown above.

(i) $H(s) = \dfrac{s+5}{s^2+2s+3}$

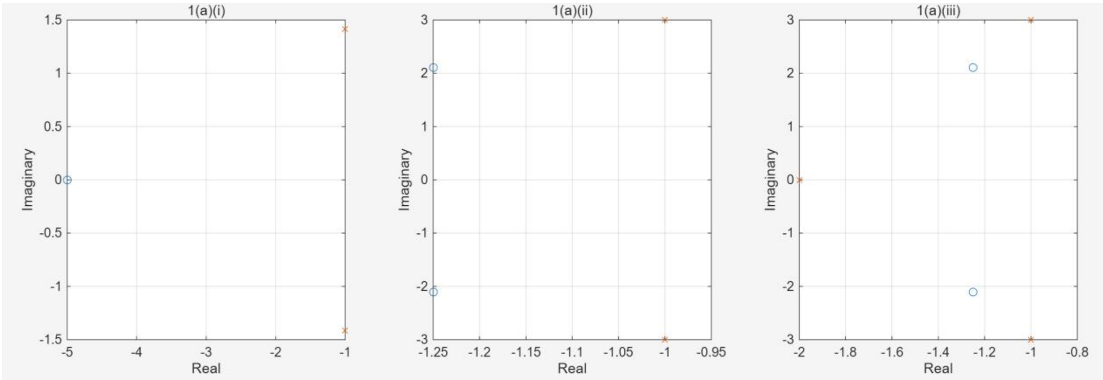
(ii) $H(s) = \dfrac{2s^2+5s+12}{s^2+2s+10}$

(iii) $H(s) = \dfrac{2s^2+5s+12}{(s^2+2s+10)(s+2)}$

Code:

```
clear; clc;
figure;

b1 = [1 5];
a1 = [1 2 3];
zs1 = roots(b1);
ps1 = roots(a1);

subplot(1, 3, 1);
plot(real(zs1), imag(zs1), 'o');
hold on
plot(real(ps1), imag(ps1), 'x');
grid on;
title('1(a)(i)');
xlabel('Real'); ylabel('Imaginary');

b2 = [2 5 12];
a2 = [1 2 10];
zs2 = roots(b2);
ps2 = roots(a2);

subplot(1, 3, 2);
plot(real(zs2), imag(zs2), 'o');
hold on
plot(real(ps2), imag(ps2), 'x');
grid on;
title('1(a)(ii)');
xlabel('Real'); ylabel('Imaginary');

b3 = [2 5 12];
a3 = conv([1 2 10], [1 2]);
zs3 = roots(b3);
ps3 = roots(a3);

subplot(1, 3, 3);
plot(real(zs3), imag(zs3), 'o');
hold on
plot(real(ps3), imag(ps3), 'x');
grid on;
title('1(a)(iii)');
xlabel('Real'); ylabel('Imaginary');
```

Output:



By using the roots function in MATLAB, we successfully calculated the poles and zeros of three continuous-time systems and plotted them on the complex plane using the plot function.We can see that for 1(a)(i), $z_0 =- 5,\ p_0 =- 1 \pm j1.414$. for 1(a)(ii), $z_0 =- 1.25\ \pm j2.107,\ p_0 = - 1 \pm j3$. for 1(a)(iii), $z_0 =- 1.25\ \pm j2.107,\ p_0 =- 1 \pm j3,\ - 2$.

3.1.b

(b). For each of the rational expressions in Part (a), determine the ROC corresponding to the stable system.

To determine if a continuous-time LTI system is stable, we need to check if its ROC includes the imaginary axis.

For system (i): The poles are $\mathrm{p} =- 1 \pm \mathrm{j}\sqrt{2}$. The real part of the poles is -1. For the ROC to include the imaginary axis, it must be on the right side of the poles. Therefore, the ROC for the stable system is $\Re\mathrm{e}(\mathrm{s}) >- 1$.

For system (ii): The poles are $\mathrm{p} =- 1 \pm 3\mathrm{j}$. The real part is also -1. Similarly, to include the imaginary axis, the ROC for the stable system must be $\Re\mathrm{e}(\mathrm{s}) >- 1$.

For system (iii): The poles are $\mathrm{p} =- 2$ and $\mathrm{p} =- 1 \pm 3\mathrm{j}$. The maximum real part of the poles here is -1. The ROC cannot contain any poles, and to be stable, it must include the imaginary axis. So, the ROC must be to the right of the rightmost pole. Therefore, the ROC is $\Re\mathrm{e}(\mathrm{s}) >- 1$.

Conclusion: For all three systems to be stable, their ROC must be $\Re\mathrm{e}(\mathrm{s}) >- 1$. Also, because the ROC extends to the right, it means the corresponding stable systems are all causal systems.

3.1.c

(c). For the causal LTI system whose input and output satisfy the following differential equation

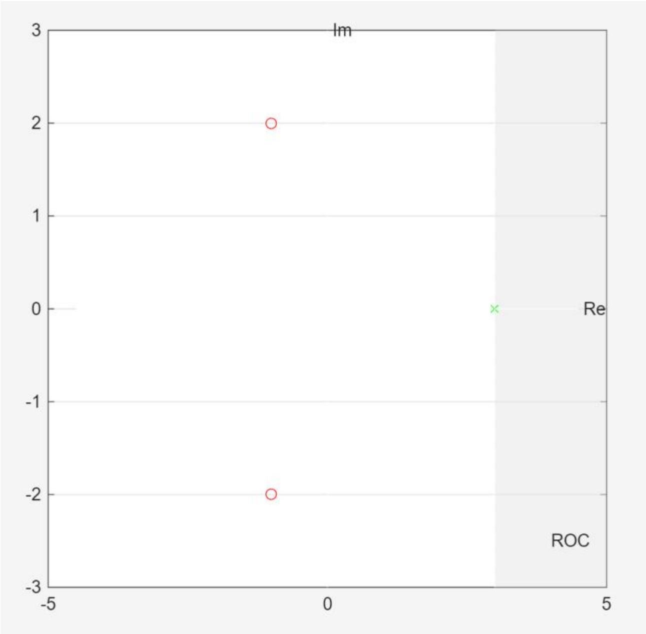
$$\frac{dy(t)}{dt} - 3y(t) = \frac{d^2x(t)}{dt^2} + 2\frac{dx(t)}{dt} + 5x(t),$$

find the poles and zeros of the system and make an appropriately labeled pole-zero diagram.

Code:

```
clear; clc;

b = [1 2 5];
a = [1 -3];
[ps, zs] = pzplot(b,a,4);
% --- 下面是手动修补阴影的代码 ---
hold on;
yl = ylim;  xl = xlim;

fill([3, xl(2), xl(2), 3], [yl(1), yl(1), yl(2), yl(2)], [0.9 0.9 0.9], ...
    'FaceAlpha', 0.5, 'EdgeColor', 'none');

uistack(findobj(gca, 'Type', 'Patch'), 'bottom');
```

Output:



The drawing logic of the pzplot function is to find the region that contains the given test point without crossing any poles. Problem 1(c) states at the beginning that this is a causal LTI system. According to signals and systems theory, the ROC of a causal continuous-time system must be to the right of the rightmost pole. In this case, the only pole is at $s = 3$, so the correct ROC for the causal system must be $\Re e(s) > 3$.

The key part of the code is the third parameter "4" in pzplot(b,a,4). This parameter acts as a "test point", telling the function that we need the ROC that contains the point $s = 4$. Since $s = 4$ is to the right of the pole $s = 3$, the function correctly draws the shaded area for $\Re e(s) > 3$, which represents a causal system.

3.1.d

> (d) Examine how **pzplot** can determine the region of convergence of a rational transform from knowledge of a single point within the ROC.

When we provide a "single point" as the third parameter to pzplot, the function uses the following logic:

First, it solves the denominator polynomial to locate all the pole "walls".

Then, it looks for the specific region that contains our provided "single point".

Finally, it extends to the left and right from that point until it hits the nearest poles (or infinity). This determines the left and right boundaries, thus finding the complete ROC.

Conclusion: Because the poles have already divided the complex plane into a few fixed regions, we only need to provide any single point inside the ROC to "uniquely identify" and automatically draw the target Region of Convergence among these candidate regions.

3.2 Problem 2

Problem 2

- (a). Use **dpzplot** to plot the poles and zeros for $H(z)$ in Eq. (1).
- (b). Use **dpzplot** to plot the poles and zeros for a filter which satisfies the difference equation

$$y[n] + y[n-1] + 0.5y[n-2] = x[n]. \tag{3}$$

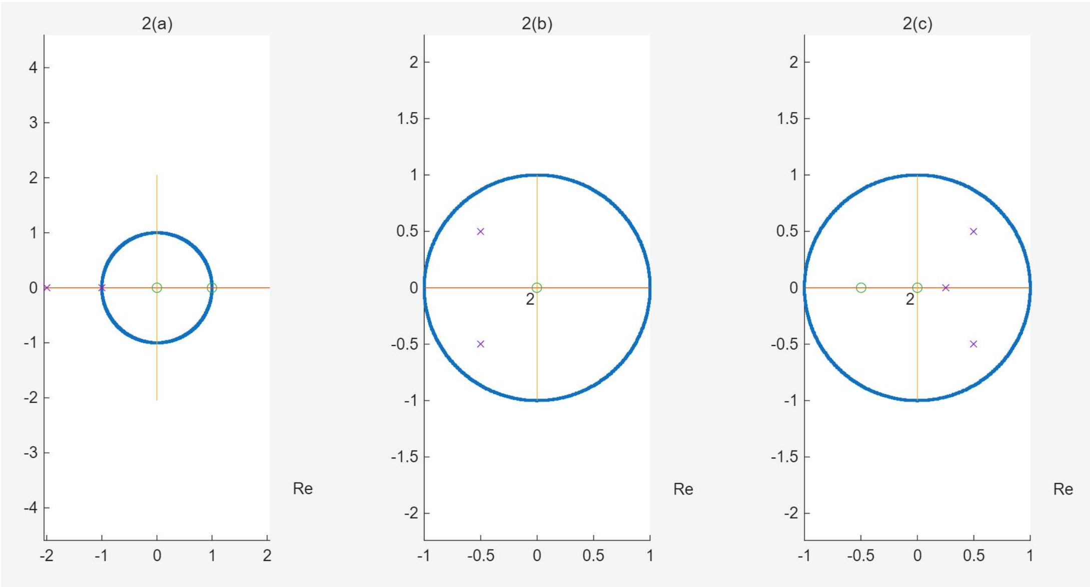
- (c). Use **dpzplot** to plot the poles and zeros for a filter which satisfies the difference equation

$$y[n]-1.25y[n-1]+0.75y[n-2]-0.125y[n-3] = x[n]+0.5x[n-1]. \tag{4}$$

Code:

```
clear; clc;

b2a = [1 -1];
a2a = [1 3 2];
subplot(1, 3, 1);
dpzplot(b2a, a2a);
title('2(a)');

b2b = 1;
a2b = [1 1 0.5];
subplot(1, 3, 2);
dpzplot(b2b, a2b);
title('2(b)');

b2c = [1 0.5];
a2c = [1 -1.25 0.75 -0.125];
subplot(1, 3, 3);
dpzplot(b2c, a2c);
title('2(c)');
```

Output:



For Figure 2(a), the poles are at $z = -1$ and $z = -2$, and the zeros are at $z = 0$ and $z = 1$.

For Figures 2(b) and 2(c), because the original numerator polynomials had a lower order than the denominators, the dpzplot function automatically padded them with zeros. This is perfectly reflected in the plots: at the origin $z = 0$ of both figures, there is a green circle marked with the number "2". This means there are two overlapping zeros at the origin. The plots also clearly show their respective complex conjugate poles.

For discrete-time systems, to determine if a causal LTI system is stable, we must check if all its poles are located strictly inside the unit circle.

Figure 2(a), the system has one pole outside the unit circle and one pole on the unit circle. Therefore, it is an unstable system.

Figure 2(b), the two poles are inside the unit circle. Therefore, if it is a causal system, it is stable.

Figure 2(c), all poles are enclosed within the unit circle, making it a stable system as well.

3.3 Problem 3

3.3.a

(a). If $H_{\mathrm{ac}}(s)$ is the system function associated with Eq. (9.12) and $H_{\mathrm{c}}(s)$ is the system function associated with Eq. (9.11), how are the poles of these two system functions related? How is $H_{\mathrm{c}}(s)$ related to $H_{\mathrm{ac}}(s)$?

According to the properties of the Laplace transform, if we time-reverse the original system, the variable s in the s-domain will also become $-s$.Therefore, the relationship between the two system functions is $H_{ac}(s) = H_c(-s)$. Since $H_{ac}(s) = H_c(-s)$, the poles of $H_{ac}(s)$ are the negative of the poles of $H_c(s)$. They are mirrored across the origin in the complex plane.

3.3.b

For the following problems, consider the differential equation

$$\frac{dy(t)}{dt} + 2y(t) = x(t)\,. \tag{9.13}$$

(b). Determine the system function $H(s)$ associated with Eq. (9.13) and all of the possible regions of convergence of $H(s)$. For each region of convergence, determine the impulse response of the corresponding LTI system.

By taking the Laplace transform of both sides of the differential equation $\frac{dy(t)}{dt} + 2y(t) = x(t)$, we get $sY(s) + 2Y(s) = X(s)$. Thus, the system function is:

$$H(s) = \frac{1}{s+2}$$

This system has a single pole at $s = -2$.

①ROC 1: $\Re e(s) > -2$

This region extends to the right, corresponding to a causal system. Taking the inverse transform gives the impulse response:

$$h(t) = e^{-2t}u(t)$$

②ROC 2: $\Re e(s) < -2$

This region extends to the left, corresponding to an anticausal system. Taking the inverse transform gives the impulse response:

$$h(t) = -e^{-2t}u(-t)$$

3.3.c

(c). For each region of convergence determined in Part (b), what is the corresponding auxiliary condition for the differential equation.

For $\Re e(s) > -2$, the corresponding auxiliary condition is Initial Rest. This means if the input signal is zero before a certain time t_0, the system's output must also be zero before t_0.

For $\Re e(s) < -2$, the corresponding auxiliary condition is Final Rest. This means if the input signal is zero after a certain time t_1, the system's output must also be zero after t_1.

3.3.d

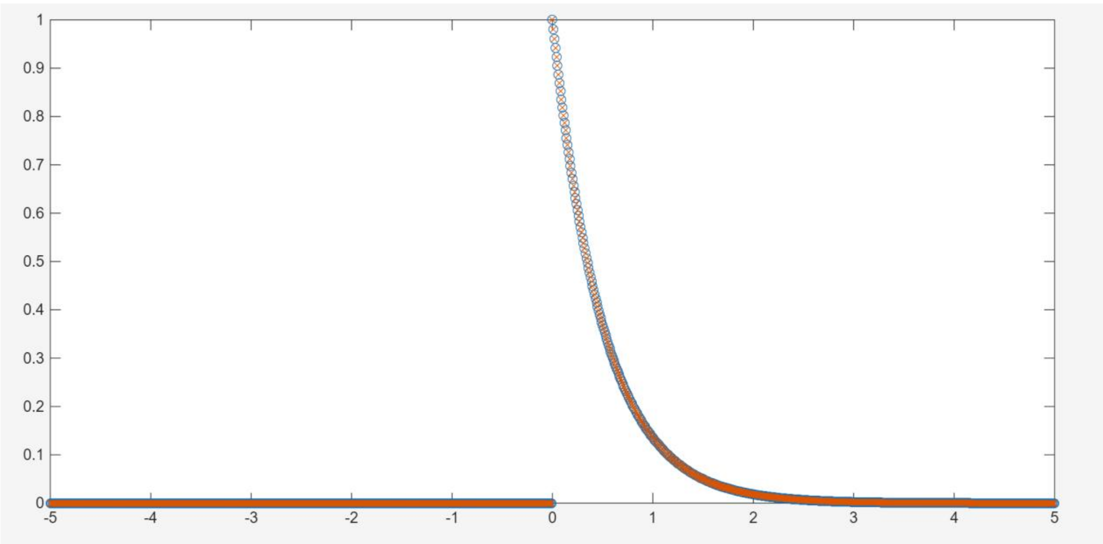
(d). Define `a` and `b` to contain the coefficients of the denominator and numerator polynomials of $H(s)$. For the causal system which has system function $H(s)$, use `impulse` to verify the analytic expression for the impulse response. Store in the vector `h` the values of the impulse response at the times in `t=[-5:0.01:5]` returned by `impulse`. Plot `h` versus `t`. Note that since `hs=impulse(b,a,ts)` returns samples of the response to $\delta(t-\texttt{ts(1)})$, you will have to select the appropriate time samples to input to `impulse` and append the result to an appropriate number of zeros.

Code:

```
clear; clc;

a = [1 2];
b = 1;

t = -5:0.01:5;

h_ana = exp(-2*t) .* (t >= 0);

t1 = 0:0.01:5;
h = impulse(b, a, t1);

hc = [zeros(length(t)-length(h), 1); h];

figure;
plot(t, h_ana, 'o');
hold on;
plot(t, hc, 'x');
```

In the previous section 3(b), we theoretically derived that the causal stable impulse response for the system function $H(s) = \frac{1}{s+2}$ is $h(t) = e^{-2t}u(t)$. In this code, we first calculated this analytic solution h_ana. Next, we used MATLAB's built-in impulse function for numerical simulation. Since impulse defaults to simulating causal systems (where the system is stimulated exactly at $t = 0$), following the manual's tip, we only passed the positive time interval t_pos = 0:0.01:5 to the function. We then manually padded the front of the resulting array with zeros to match the full plotting time axis [-5, 5].

Output:



Conclusion: From the generated graph, it is clear that the 'o' markers and 'x' markers overlap perfectly. In the interval t < 0, the impulse response is strictly 0; in the interval $\mathrm{t} \leq 0$, both show the standard e^{-2t} exponential decay curve. This perfectly verifies that our theoretical derivation in 3(b) is correct. It also proves that the method of "calculating for the positive half-axis and padding zeros for the negative half-axis" is an effective and accurate way to use the impulse function to plot a causal system's response over a complete time axis that includes negative time.

3.3.e

(e). Repeat Part (d) for the anticausal system. Remember that **impulse(b,a,ts)** assumes that the coefficients in **a** and **b** correspond to a causal system. You will need to define a new set of coefficients for a time-reversed system (see Eq. (9.12)) and then appropriately flip the impulse response computed by **impulse** (since the output of Eq. (9.12) is time-reversed).

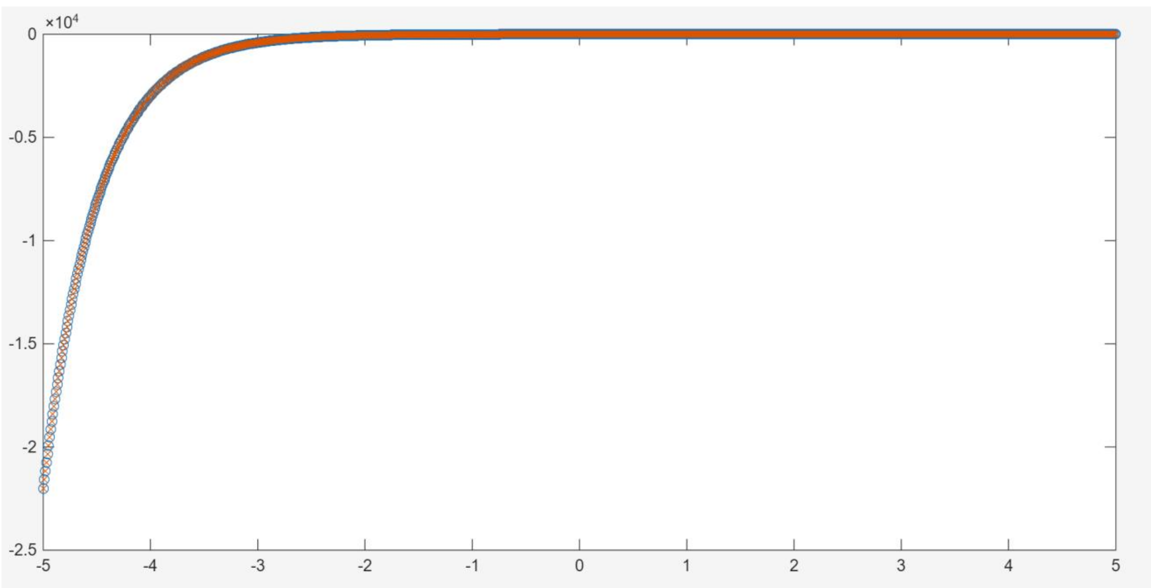
Code:

```
clear; clc;

a = [-1 2];
b = 1;

t = -5:0.01:5;

h_ana = -exp(-2*t) .* (t <= 0);

t1 = 0:0.01:5;
h = impulse(b, a, t1);

hc = [zeros(length(t)-length(h), 1); h];
hc = flip(hc);

plot(t, h_ana, 'o');
hold on;
plot(t, hc, 'x');
```

According to Eq. (9.12) provided in the PPT, by replacing t with $-t$ in the original differential equation $\frac{dy(t)}{dt} + 2y(t) = x(t)$, the coefficient of the first-derivative term changes sign, resulting in the new coefficients a_new = [-1 2] and b_new = 1.

we first padded a zero vector corresponding to the negative time length to the front of the positive-axis response, and then used the flip function to reverse the entire array. This not only "moved" the response to the negative time axis but also simultaneously completed the mirror flip of the time axis, thus obtaining the true anticausal impulse response.

Output:



Conclusion: The generated graph shows that the 'o' markers representing the analytical solution and the 'x' markers representing the simulated solution overlap perfectly over the entire time interval of [-5, 5]. The response is strictly zero when $t > 0$, and shows a reversed exponential growth curve when $t \leq 0$.

3.3.f

(f). Analytically calculate the output of the anticausal LTI system satisfying Eq. (9.13) when the input is $x(t) = e^{5t/2}u(-t)$.

Given the input signal $x(t) = e^{2.5t}u(-t)$ (which is $e^{5t/2}$). According to Laplace transform pairs, its transform is:

$$X(s) = \frac{-1}{s - 2.5}$$

The system function for the differential equation $\frac{dy(t)}{dt} + 2y(t) = x(t)$ is:

$$H(s) = \frac{1}{s + 2}$$

The output is $Y(s) = H(s)X(s)$:

$$Y(s) = \left(\frac{1}{s+2}\right)\left(\frac{-1}{s-2.5}\right) = \frac{-1}{(s+2)(s-2.5)}$$

The ROC of Y(s) is the intersection of the ROCs of X(s) and H(s): $\Re e(s) < 2.5 \cap \Re e(s) <- 2$. Therefore, the ROC for Y(s) is $\Re e(s) <- 2$.

$$Y(s) = \frac{2/9}{s+2} - \frac{2/9}{s-2.5}$$

we get the final time-domain output:

$$y(t) = \frac{2}{9}(-e^{-2t}u(-t)) - \frac{2}{9}(-e^{2.5t}u(-t))$$

$$y(t) = \frac{2}{9}(e^{2.5t} - e^{-2t})u(-t)$$

3.3.g

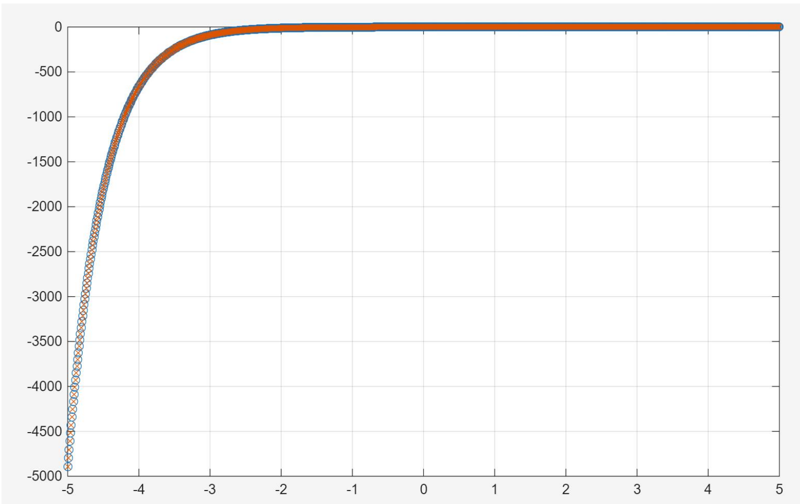
(g). Use `lsim` to verify the output of the anticausal system derived in Part (f) at the time samples `t`. Like `impulse`, the function `lsim(b,a,x,ts)` also assumes that the vectors `a` and `b` correspond to a causal system, so you must work with the time-reversed differential equation. The coefficients of this differential equation should already have been computed in Part (e). Note that the input to the time-reversed system must be time-reversed, i.e., $r(t) = x(-t)$.

Code:

```
clear; clc;

t = -5:0.01:5;
t1 = 0:0.01:5;

y_ana = (2/9) * (exp(2.5*t) - exp(-2*t)) .* (t <= 0);

a_new = [-1 2];
b_new = 1;

r = exp(-2.5*t1);
w = lsim(b_new, a_new, r, t1);
lw = length(w);

y_sim = [zeros(length(t)-lw, 1); w];
y_sim = flip(y_sim);

figure;
plot(t, y_ana, 'o');
hold on;
plot(t, y_sim, 'x');
grid on;
```

we time-reversed the original system's differential equation to obtain the causal system coefficients a_new = [-1 2]. Second, the input signal must also be time-reversed, forming $r(t) = x(-t) = e^{-2.5t}u(t)$. This turns $r(t)$ into a standard right-sided causal signal. We fed the reversed system and signal into $lsim(b_n ew, a_n ew, r, t1)$ to calculate the response on the positive half-axis. Finally, similar to 3(e), we padded the output with zeros and used the flip function to reverse the entire result along the time axis again, restoring the true anticausal system output.

Output:



Conclusion: The generated plot shows that the 'o' markers (theoretical analytical solution) and the 'x' markers (lsim simulated solution) match perfectly. In the region where t > 0, the output is strictly 0; in the region where $t \leq 0$, the curve displays the dynamic behavior of $\frac{2}{9}(e^{2.5t} - e^{-2t})$.

This perfect overlap confirms two important facts: First, our mathematical derivation using Laplace partial fraction expansion in 3(f) is correct. Second, the three-step strategy of simultaneously reversing the system equation and the input signal, and then reversing the final result back, can extremely accurately utilize functions originally designed only for causal systems (like lsim) to simulate the forced response of anticausal systems.

3.3.h

Now consider the third-order differential equation

$$\frac{d^3y(t)}{dt^3} + \frac{d^2y(t)}{dt^2} + 24\,\frac{dy(t)}{dt} - 26\,y(t) = \frac{d^2x(t)}{dt^2} + 7\,\frac{dx(t)}{dt} + 21\,x(t)\,. \qquad (9.14)$$

(h). Determine the system function $H(s)$ associated with Eq. (9.14) and plot the poles and zeros of this system function as shown in Tutorial 9.1. Determine all of the possible regions of convergence. For which region of convergence is the system stable?

Taking the Laplace transform of both sides of the given third-order differential equation, we extract the system function H(s):

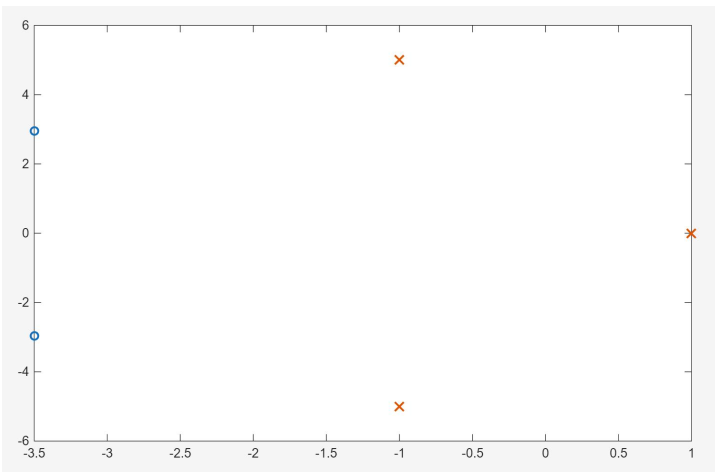
$$H(s) = \frac{s^2 + 7s + 21}{s^3 + s^2 + 24s - 26}$$

Code:

```
clear; clc;

b_3rd = [1 7 21];
a_3rd = [1 1 24 -26];

zs_3rd = roots(b_3rd);
ps_3rd = roots(a_3rd);

disp('Zeros of H(s):'); disp(zs_3rd);
disp('Poles of H(s):'); disp(ps_3rd);

figure;
plot(real(zs_3rd), imag(zs_3rd), 'o', 'LineWidth', 1.5);
hold on;
plot(real(ps_3rd), imag(ps_3rd), 'x', 'MarkerSize', 10, 'LineWidth', 1.5);
```

Output:

```
命令行窗口
Zeros of H(s):
  -3.5000 + 2.9580i
  -3.5000 - 2.9580i

Poles of H(s):
  -1.0000 + 5.0000i
  -1.0000 - 5.0000i
   1.0000 + 0.0000i
```



By using MATLAB to calculate the roots of its numerator and denominator, we obtain:

Zeros: $z = -3.5 \pm 2.958j$

Poles: $s = 1$ and $s = -1 \pm 5j$

These three poles divide the complex plane into three strips. Three possible ROCs:

①$\Re\mathrm{e}(s) > 1$ ②$-1 < \Re\mathrm{e}(s) < 1$ ③$\Re\mathrm{e}(s) < -1$

The absolute criterion for a continuous-time LTI system to be stable is: its ROC must include the imaginary axis (where $\Re\mathrm{e}(s) = 0$). Only the region $-1 < \Re\mathrm{e}(s) < 1$ contains the imaginary axis where $\Re\mathrm{e}(s) = 0$.

Therefore, we conclude: The system is stable if and only if the region of convergence is $-1 < \Re\mathrm{e}(s) < 1$. It is a noncausal system.

3.3.i

(i). Use **residue** to determine the partial fraction expansion of $H(s)$. For each region of convergence determined in Part (h), analytically determine the associated impulse response.

Code:

```
clear; clc;

b_3rd = [1 7 21];
a_3rd = [1 1 24 -26];

[r, p, k] = residue(b_3rd, a_3rd);

disp('Residues (r):');
disp(r);
disp('Poles (p):');
disp(p);
disp('Direct Term (k):');
disp(k);
```

Output:

```
命令行窗口
Residues (r):
   0.0000 - 0.5000i
   0.0000 + 0.5000i
   1.0000 + 0.0000i

Poles (p):
  -1.0000 + 5.0000i
  -1.0000 - 5.0000i
   1.0000 + 0.0000i

Direct Term (k):
>>
```

The residue function outputs the following:

Poles p: 1 and $-1 \pm 5j$

Corresponding residues r: 1 and $\mp 0.5j$

Direct term k: empty [].

This means the system function $H(s)$ can be expanded as:

$$H(s) = \frac{1}{s-1} + \frac{-0.5j}{s-(-1+5j)} + \frac{0.5j}{s-(-1-5j)}$$

To make the inverse Laplace transform easier, we combine the two complex conjugate terms:

$$H(s) = \frac{1}{s-1} + \frac{5}{(s+1)^2 + 5^2}$$

Now we analytically find the impulse response for each ROC:

ROC 1: $\Re\mathrm{e}(s) > 1$ (Causal System)

This ROC is to the right of all poles. Therefore, all partial fraction terms correspond to right - sided (causal) signals. Using the Laplace transform pairs $\mathcal{L}^{-1}\frac{1}{s-a} = e^{at}u(t)$ and $\mathcal{L}^{-1}\frac{\omega}{(s+\alpha)^2+\omega^2} = e^{-\alpha t}\sin(\omega t)u(t)$, we get:

$$h_1(t) = e^t u(t) + e^{-t}\sin(5t)u(t)$$

ROC 2: $-1 < \Re\mathrm{e}(s) < 1$ (Stable, Noncausal System)

This ROC is to the left of the pole $s = 1$, but to the right of the poles $s = -1 \pm 5j$. Thus, the pole at 1 corresponds to a left - sided signal, and the poles at $-1 \pm 5j$ correspond to a right-sided signal.

$$h_2(t) = -e^t u(-t) + e^{-t}\sin(5t)u(t)$$

ROC 3: $\Re\mathrm{e}(s) < -1$ (Anticausal System)

This ROC is to the left of all poles. Therefore, all terms correspond to left - sided (anticausal) signals.

$$h_3(t) = -e^t u(-t) - e^{-t}\sin(5t)u(-t)$$

3.3.j

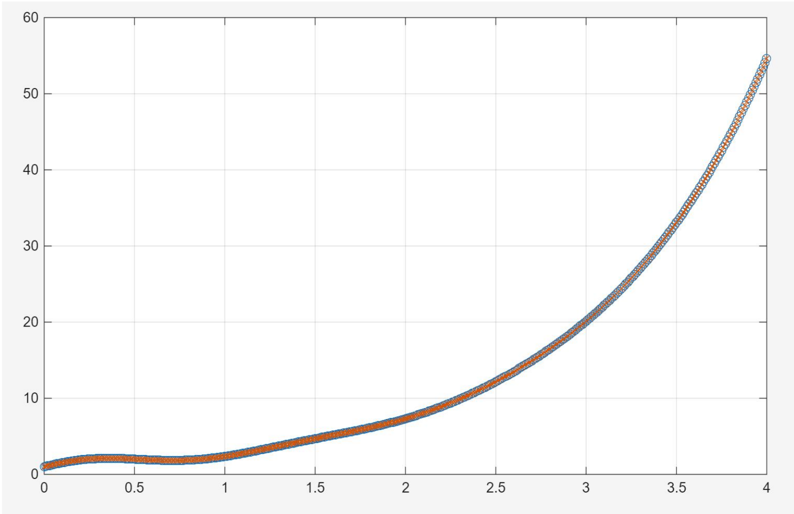
(j). For the causal system associated with Eq. (9.14), use **impulse** to verify the corresponding impulse response computed in Part (i). Use the vector **t** for the time samples. What are the auxiliary conditions on $y(t)$ associated with this causal system?

Code:

```
clear; clc;

b_3rd = [1 7 21];
a_3rd = [1 1 24 -26];

t = 0:0.01:4;

h_ana = exp(t) + exp(-t) .* sin(5*t);
h_sim = impulse(b_3rd, a_3rd, t);

figure;
plot(t, h_ana, 'o');
hold on;
plot(t, h_sim, 'x');
grid on;
```

Analysis: In 3(i), we derived the causal impulse response for the ROC $\Re\mathrm{e}(s) > 1$ as $h_1(t) = e^t u(t) + e^{-t}\sin(5t)u(t)$. In this code, we plotted this pure mathematical analytical expression (blue circles 'o') against the simulation result directly obtained using MATLAB's impulse function (red crosses 'x') for comparison.

Output:



Auxiliary Conditions: Initial Rest. For an LTI system described by a linear constant-coefficient differential equation, to make it a causal system, it must satisfy the initial rest condition. $y(0^-) = 0, y'(0^-) = 0, y''(0^-) = 0.$

Conclusion: The generated graph shows that the analytical solution and the numerical simulation solution overlap perfectly, re-verifying the correctness of our partial fraction expansion in 3(i). In addition, from the trend of the curve growing exponentially over time, we can visually see that although the system satisfies the causal condition of initial rest, it is an extremely unstable system due to the pole in the right half-plane s = 1 causing the e^t term to diverge.

3.3.k

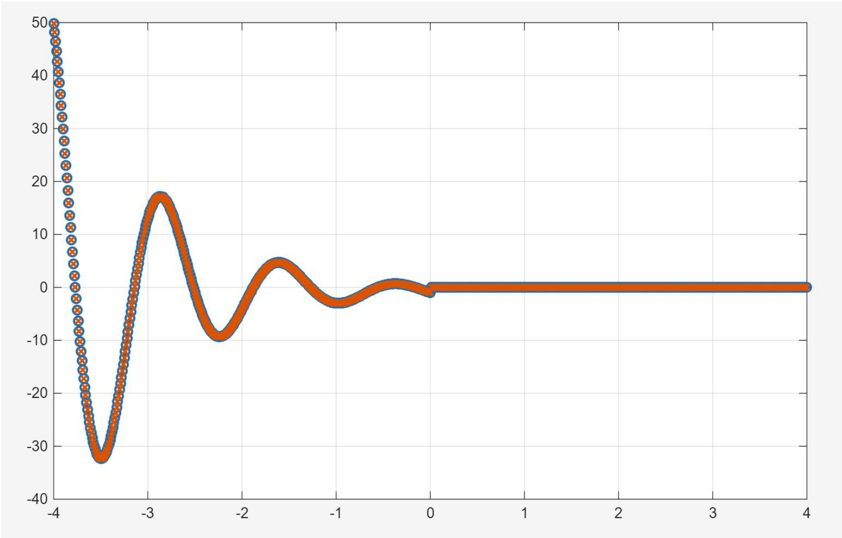
(k). For the anticausal system associated with Eq. (9.14), use **impulse** to verify the corresponding impulse response computed in Part (i). Use the vector **t** for the time samples. Remember that you must first determine the coefficients of the time-reversed differential equation, and then derive the anticausal impulse response of Eq. (9.14) from the causal impulse response of the time-reversed equation. What are the auxiliary conditions on $y(t)$ associated with this anticausal system?

Code:

```
clear; clc;

t = -4:0.01:4;
t_pos = 0:0.01:4;

h_ana = ( -exp(t) - exp(-t).*sin(5*t) ) .* (t <= 0);

a_rev = [-1  1 -24 -26];
b_rev = [ 1 -7  21];

h_rev_pos = impulse(b_rev, a_rev, t_pos);

lh = length(h_rev_pos);
h_sim = [zeros(length(t)-lh, 1); h_rev_pos];
h_sim = flip(h_sim);

figure;
plot(t, h_ana, 'o', 'LineWidth', 1.5);
hold on;
plot(t, h_sim, 'x', 'LineWidth', 1.5);
grid on;
```

Analysis: In 3(i), we derived that when the ROC is $\Re e(s) <- 1$, the third-order system is an anticausal system with the analytical impulse response $h_3(t) =- e^t u(-t) - e^{-t}\sin(5t)u(-t)$. Since the built-in impulse function cannot directly handle anticausal systems, we had to time-reverse the high-order differential equation. According to derivative rules, the coefficients of odd-order derivatives (1st, 3rd) change signs, while even-order derivatives (0th, 2nd) keep their signs. Thus, the original coefficients $a = [1,1,24,-26]$ and $b = [1,7,21]$ become $a_{rev} = [-1,1,-24,-26]$ and $b_{rev} = [1,-7,21]$.

Output:



Auxiliary Conditions: Final Rest. $y(0^+) = 0, y'(0^+) = 0, y''(0^+) = 0$.

Conclusion: The generated simulation graph shows a perfect overlap between the theoretical and simulated values, verifying that the inverse transform calculation for the left-sided signal in 3(i) is correct. Furthermore, looking at the shape of the curve for $t \leq 0$, we can see that as time goes backwards to negative infinity, the signal not only contains intense sinusoidal oscillations but also, due to the e^{-t} term, the amplitude explodes exponentially. This visually indicates that this anticausal system contains left-half-plane poles, making it an extremely unstable system for a leftward-extending ROC.

3.3.l

The following problems consider the numerical implementation of the stable LTI system whose inputs and outputs satisfy Eq. (9.14). The impulse response $h_s(t)$ of this system, as calculated in Part (i), should be nonzero for all time t. This system is called noncausal since it is neither causal nor anticausal, and a simple time-reversal of the differential equation will not suffice for implementation. Instead, the differential equation must be decomposed into a causal and an anticausal component, each of which are computed separately. This is known as a parallel realization of the system.

(l). A parallel realization an LTI system is easily visualized by decomposing the system function as $H(s) = H_1(s) + H_2(s)$, so that the output $y(t)$ can be computed by separately computing the response to $H_1(s)$ and $H_2(s)$. Note that the regions of convergence of $H_1(s)$ and $H_2(s)$ must be properly specified. Given the partial fraction expansion you found in Part (i), determine $H_1(s)$ and $H_2(s)$ for the stable system such that $h_1(t)$ is causal and $h_2(t)$ anticausal. Make sure to specify the regions of convergence of both $H_1(s)$ and $H_2(s)$.

$$H(s) = \frac{1}{s-1} + \frac{5}{(s+1)^2 + 5^2}$$

The problem requires decomposing into $H(s) = H_1(s) + H_2(s)$, where $h_1(t)$ is causal (right-sided ROC) and $h_2(t)$ is anticausal (left-sided ROC). To keep the overall system stable, the intersection of their ROCs must be exactly $-1 < \Re e(s) < 1$.

$$H_1(s) = \frac{5}{s^2 + 2s + 26}$$

$$H_2(s) = \frac{1}{s-1}$$

$H_1(s)$ is a stable causal system with an ROC of $\Re e(s) >- 1$, and $H_2(s)$ is a stable anticausal system with an ROC of $\Re e(s) < 1$. The intersection of these two ROCs perfectly forms the ROC of the original stable system$(- 1 < \Re e(s) < 1)$.

3.3.m

(m). Determine $h_1(t)$ and $h_2(t)$, the inverse transforms of $H_1(s)$ and $H_2(s)$. You should be able to determine these impulse responses from the expression for $h_s(t)$.

The right - sided signal term corresponds to the inverse transform of the causal system $H_1(s)$:

$$h_1(t) = e^{-t} \sin(5t) u(t)$$

The left -

sided signal term corresponds to the inverse transform of the anticausal system $H_2(s)$:

$$h_2(t) =- e^t u(-t)$$

3.3.n

(n). Define $y_1(t)$ to be the output of the system defined by $H_1(s)$ and its region of convergence. Specify the differential equation corresponding to $H_1(s)$ along with the associated auxiliary conditions on $y_1(t)$.

We have:

$$H_1(s) = \frac{Y_1(s)}{X(s)} = \frac{5}{s^2 + 2s + 26}$$

$$Y_1(s)(s^2 + 2s + 26) = 5X(s)$$

$$s^2 Y_1(s) + 2sY_1(s) + 26Y_1(s) = 5X(s)$$

$$\frac{d^2 y_1(t)}{dt^2} + 2\frac{dy_1(t)}{dt} + 26y_1(t) = 5x(t)$$

Auxiliary Conditions:

$H_1(s)$ describes a causal system.

For a causal LTI system described by a differential equation, the required auxiliary condition is Initial Rest.

$$y_1(t_0^-) = 0, \quad \frac{dy_1(t_0^-)}{dt} = 0$$

3.3.o

(o). Define $y_2(t)$ to be the output of the system defined by $H_2(s)$ and its region of convergence. Specify the differential equation corresponding to $H_2(s)$ along with the associated auxiliary conditions on $y_2(t)$.

we have:

$$\mathrm{H_2(s)} = \frac{\mathrm{Y_2(s)}}{\mathrm{X(s)}} = \frac{1}{\mathrm{s-1}}$$

$$\mathrm{Y_2(s)(s-1) = X(s)}$$

$$\mathrm{sY_2(s) - Y_2(s) = X(s)}$$

$$\frac{\mathrm{dy_2(t)}}{\mathrm{dt}} - \mathrm{y_2(t) = x(t)}$$

Auxiliary Conditions:

the future state of the system after the excitation must be zero:

$$y_2(t_1^+) = 0$$

3.3.p

(p). Use **`lsim`** to determine the response of the stable system to the input

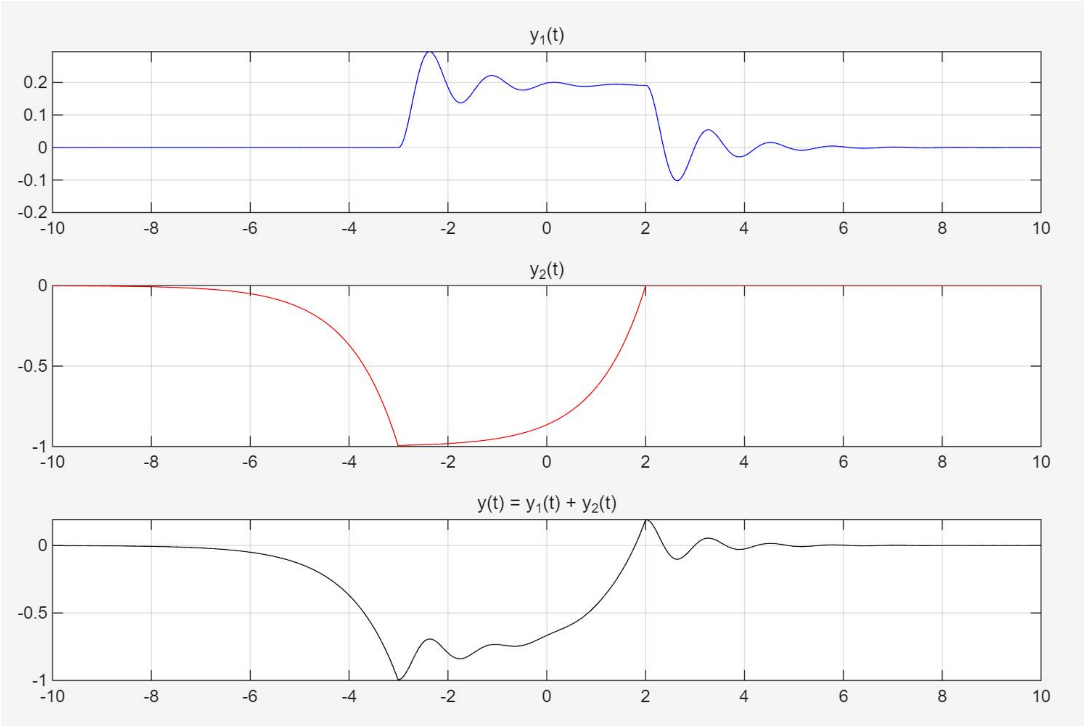
$$x(t) = \begin{cases} 1, & -3 \le t \le 2, \\ 0, & \text{otherwise}, \end{cases} \tag{9.15}$$

on the interval $-10 \le t \le 10$ for the time samples in `t=[-10:0.01:10]`. Plot $y_1(t)$, $y_2(t)$ and $y(t)$ on the interval $-10 \le t \le 10$.

Code:

```
clear; clc;

t = -10:0.01:10;
x = double((t >= -3) & (t <= 2));

b1 = 5;
a1 = [1 2 26];
y1 = lsim(b1, a1, x, t);

b2_rev = 1;
a2_rev = [-1 -1];
r = flip(x);
w = lsim(b2_rev, a2_rev, r, t);
y2 = flip(w);

y = y1 + y2;

figure;
subplot(3, 1, 1);
plot(t, y1, 'b');
grid on;
title('y_1(t)');

subplot(3, 1, 2);
plot(t, y2, 'r');
grid on;
title('y_2(t)');

subplot(3, 1, 3);
plot(t, y, 'k');
grid on;
title('y(t) = y_1(t) + y_2(t)');
```

Output:



For the causal component $y_1(t)$: We directly inputted x(t) into the system defined by $H_1(s)$. The simulation plot clearly demonstrates the "initial rest" property—the output is strictly 0 before the input signal begins $t = -3$. After the pulse ends (t>2), the output does not disappear immediately but exhibits the typical decaying oscillation (ringing) of a second-order system.

For the anticausal component $y_2(t)$: We used the double time-reversal technique. The simulation plot shows the highly typical "final rest" and noncausal predictive properties—the output is strictly 0 after the input signal ends $t = 2$; however, before the input pulse arrives ($t < -3$), the output signal already begins to show an exponential rising trend (as if the system "foresees" the upcoming pulse).

For the total response y(t): It is the superposition of $y_1(t)$ and $y_2(t)$. As seen in the bottom subplot, the total response possesses both the ability to anticipate the input (the exponential rise for $t < -3$) and the inertial continuation after the input disappears (the damped oscillation for t>2). Most importantly, over the entire time axis of [-10, 10], the signal amplitude always remains within a finite range without any divergence.

Conclusion: The experiment demonstrates that complex noncausal stable boundary conditions can be losslessly decomposed into a "causal stable process with initial rest" and an "anticausal stable process with final rest".

4 Experimental Conclusions

(1) System Stability & ROC: A system's stability depends entirely on its Region of Convergence (ROC). A continuous-time system is stable if its ROC includes the imaginary axis. A discrete-time system is stable if all its poles are inside the unit circle.

(2) Order Mismatches: In discrete systems, if the numerator's order is lower than the denominator's, MATLAB (dpzplot) automatically appends zeros at the origin to maintain

characteristic accuracy.

(3) Anticausal Simulation: Since standard functions (like impulse or lsim) only support causal systems, anticausal systems can be simulated using a "time-reversal" technique: reverse the differential equation, calculate the causal response, and then flip the output timeline.

(4)Parallel Realization: Complex stable noncausal systems can be accurately simulated by decomposing them (via partial fraction expansion) into a causal part and an anticausal part. We can simulate them separately and add the results together.